An Explainable Deep Learning Framework for Multi-Class Retinal Disease Classification Using Fundus Images

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***Abstract*— Retinal diseases are a significant contributor to vision loss and early and precise diagnosis is critical. This study introduces a deep learning approach to multi class retinal disease classification from fundus images. The proposed framework divides four types of retinal diseases: Dry Age Related Macular Degeneration, Wet Age Related Macular Degeneration, Moderate Diabetic Retinopathy and Normal Fundus. Implements the framework with the Medical Open Network for Artificial Intelligence library with transfer learning models: VGG16, ResNet50. The accuracy, precision, recall and F1 score are used to evaluate the model performance and they get a classification accuracy of 98.80% and an F1 score of 98.40% respectively. Moreover, Gradient Weighted Class Activation Mapping is integrated to improve the interpretability of the model and computer aided diagnosis reliability.**

Keywords— Deep Learning, Transfer Learning, Retinal Disease Classification, Fundus Image, VGG16, Resnet50, XAI, Gradient Weighted Class Activation Marking, Computer Aided Diagnosis

I. INTRODUCTION

Retinal diseases such as Age Related Macular Degeneration (AMD) and Diabetic Retinopathy (DR) are among the main causes of irreversible vision impairment and blindness worldwide, thus early diagnosis is vital to maintain vision and optimized patient care [1]. In recent years, the automated analysis of retinal fundus images has become more accurate due to the advances in artificial intelligence, especially deep learning, which has led to the development of computer aided diagnosis systems that assist ophthalmologists in the diagnosis of retinal diseases with high accuracy [2].

Transfer learning is a technique that has seen large adoption in retinal image classification as it is able to learn the discriminative retinal features without requiring extensive computation and training time by using pre trained convolutional neural networks like VGG16 and Resnet50 [3]. Although the predictive power of deep learning models is high, they are often a black box model that cannot be used in clinical practice. The techniques of Explanation of Artificial Intelligence (EAI) have the potential to make models more transparent and trusted to clinicians, as they can give visual explanations by showing the retinal regions that are most important to model predictions, such as Gradient Weighted Class Activation Mapping (Grad CAM) [4].

A deep learning framework for multi class retinal disease classification with retinal fundus images is proposed. The framework is put in place with the Medical Open Network for Artificial Intelligence (MONAI) and uses transfer learning by using VGG16 and ResNet50 to classify Dry Age Related Macular Degeneration, Wet Age Related Macular Degeneration, Moderate Diabetic Retinopathy, and Normal Fundus. Model performance is assessed with the accuracy, precision, recall, F1 score, ROC AUC and confusion matrix, and Grad CAM is introduced to enhance the interpretability of the model and aid in reliable computer aided diagnosis [5].

A. Aims and Objectives

To design an explainable deep learning framework to automatically classify retinal diseases based on fundus image classification with transfer learning models using the Medical Open Network for Artificial Intelligence (MONAI) to aid in the accurate and interpretable computer-aided diagnosis of retinal disease.

B. Objectives

- To create a deep learning system for the classification of retinal diseases in multiple classes from fundus images.
- Utilize pre-processing and data augmentation for images within MONAI.
- To implement transfer learning with VGG16 and ResNet50 for retinal diagnosis of diseases.
- To categories retinal images into normal fundus moderate diabetic retinopathy, Dry AMD and wet AMD.
- To assess model performance by accuracy, precision, recall, F1 score, ROC AUC and confusion matrix. Compare results of VGG16 and ResNet50 for the classification task.
- To implement Gradient Weighted Class Activation Mapping (Grad-CAM) to generate visual explanations of the model predictions and increase interpretability.

In this study, the automated classification of retinal diseases via deep learning framework using fundus images is focused. Image pre-processing, data augmentation, transfer learning with VGG16 and ResNet50, performance evaluation and explainability with Gradient Weighted Class Activation Mapping. The study categorizes four retinal conditions: Dry Age Related Macular Degeneration, Wet Age Related Macular Degeneration, Moderate Diabetic Retinopathy and Normal Fundus. The literature review presented in Section II addresses the classification of retinal diseases, the use of deep learning, the transfer learning approach, and explainable artificial intelligence. In Section III, the dataset, pre-processing procedure, framework of MONAI, architecture of the model, and training procedure are described. The experimental results, performance evaluation, comparative analysis and visualizations using Grad CAM are provided in Section IV. Finally, Section V wraps up the study and indicates future research avenues.

II. LITERATURE REVIEW

A systematic review of 18 studies with over 778,000 retinal images showed that deep learning models such as ResNet, VGG, DenseNet, AlexNet, and CapsNet have high diagnostic accuracy in Age Related Macular Degeneration (AMD). The results showed the effectiveness of automated AMD detection using CNN [6].

AI algorithms for AMD diagnosis have advanced from conventional machine learning algorithms based on handcrafted features to sophisticated deep learning algorithms on fundus photography, Optical Coherence Tomography (OCT), and OCT Angiography (OCTA). The need of publicly available retinal datasets and transfer learning in enhancing diagnostic performance was also emphasized [7].

Fundus photography, OCT and OCTA images have been used to demonstrate the potential of AI in the diagnosis of AMD, prediction of disease progression and evaluation of treatment response. Clinically important requirements that were identified for clinical adoption were improved interpretability and better model generalization [8].

Deep learning methods have shown to be effective in detecting, grading, predicting prognosis, and predicting treatment for AMD in multiple retinal imaging modalities. But, clinical validation and explainable artificial intelligence are still necessary for ensuring reliable deployment in healthcare settings [9].

Using advanced deep learning models, such as convolutional neural networks (CNN), transfer learning, and multimodal learning, have enhanced the evaluation and monitoring of AMD progression. One of the current research challenges is the availability of standardized datasets and external validation [10].

Retinal imaging has been used to detect and segment the Geographic Atrophy associated with AMD with good performance by deep learning models. While results are encouraging, there are important limitations including limited external validation and real world clinical implementation [11].

Combining multiple convolutional neural network (CNN) architectures has been shown to outperform individually-learned models for dry AMD classification from OCT images [12]. Feature representation fusion of complementary features had positive effect on classification accuracy and robustness.

Recent deep learning methods have been successful at detecting AMD from retinal images and identifying disease progression of AMD. However, there are some challenges that hinder the clinical use of these models, including the diversity of datasets, the lack of generalization capabilities, and the lack of explainability [13].

Combining OCT, OCTA, and fundus images has been shown to improve the performance for the detection of intermediate dry AMD, compared to single-modality methods. Multiple imaging modalities offer more complete representation of retinal features by integrating them together [14].

Fundus and OCT images have been used to effectively classify retinal diseases with the use of deep learning models, such as convolutional neural networks, vision transformers, and hybrid models. Existing studies focus on the importance of explaining, generalizing and clinically deploying AI systems [15].

Multimodal deep learning approaches that combine OCT, OCTA, fundus photography, and three-dimensional retinal scans have enhanced the classification of late-stage AMD and found clinically relevant retinal biomarkers similar to expert ophthalmologists [16].

Conventional grading techniques have proven to be less effective compared to deep learning techniques in predicting the onset and progression of AMD. To enhance clinical reliability, future research should base its work on standardized dataset, robust external validation and explainable AI. MONAI Generative Models is a standardized framework for implementing medical imaging generative AI [17].

It is designed on a modular basis and enables image generation, augmentation, anomaly detection and generation of synthetic data, which allows for reproducible development of deep learning applications [18].

III. METHODOLOGY

A deep learning framework using transfer learning and explainable artificial intelligence (XAI) is applied to classify four retinal diseases using retinal fundus images. The methodology involves data preprocessing using MONAI, partitioning of the dataset, feature learning using VGG16 and ResNet50 networks, performance evaluation, and visual interpretation using the Grad-CAM. A systematic workflow is followed that is reliable and interpretable for retinal disease classification, as depicted in the conceptual framework in Fig. 1 below.

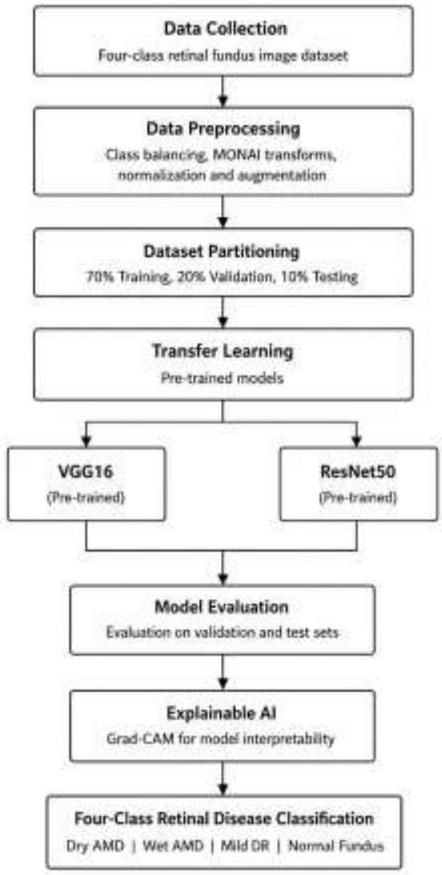

Figure 1: Proposed Methodology

A. Data Collection

The retinas were taken from the Kaggle retinal fundus image repository known as "Retinal Final Dataset" [19]. The original data set contains nine categories of retinal diseases gathered from fundus photographs. For the purpose of this research, four clinically relevant classes were chosen: Dry Age Related Macular Degeneration (Dry AMD), Wet Age Related Macular Degeneration (Wet AMD), Moderate Diabetic Retinopathy (Moderate DR), and Normal Fundus. The other four classes were omitted to build a focused multi class classification framework. The input for the proposed deep learning framework consists of retinal fundus images of each selected class, which consist of the retinal condition of each class. The chosen images were merged into a single dataset and then class balancing was applied to minimize class imbalance and to create a relatively equal distribution of samples across all four classes. This curated dataset provided a consistent foundation for the training, validation and testing of the transfer learning models.

B. Data Preprocessing

The classes in the chosen dataset were not uniformly distributed, with a large imbalance in the number of retinal fundus images in each of the four classes as shown in Fig. 2. Deep learning models can be skewed towards the majority class when the classes are imbalanced, which negatively affects the classification accuracy of the minority classes. Hence, a data balancing method was adopted before training the model to achieve uniform distribution of samples.

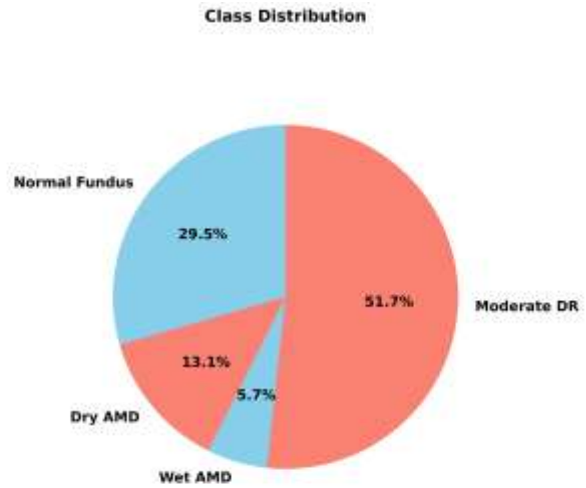

Figure 2: Imbalanced Data

The goal for data balancing was set to 1,100 images per class. Classes with over 1100 images were minimized by randomly discarding some images from the larger classes, and classes with fewer images were maximized by randomly adding images to these classes until the desired number was reached. The balancing process yielded an equal number of images for all four classes as shown in Fig. 3, which results in a balanced dataset for further pre-processing and training of the model.

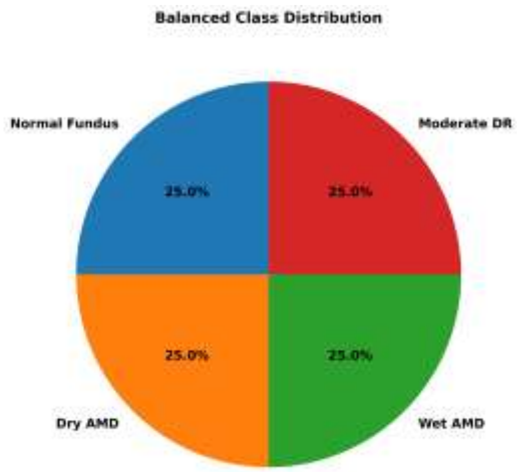

Figure 3: Balanced Data

Balanced retinal fundus image dataset was preprocessed to set a standard input for model training. MONAI was used as a tool to implement a standard pre-processing workflow in medical image analysis. Fixed input resolution for all images, and normalization of pixel intensity values. The model was trained using data augmentation methods like random horizontal flipping, random rotation, and random zoom to enhance the robustness and prevent overfitting. Processed images were then transformed to the tensor input format of the VGG16 and ResNet50 models. These pre-processing steps have helped to increase the uniformity of the image and increase the efficiency of the classification framework proposed.

C. Modeling and Benchmarking

Transfer learning is used to build the framework for the classification of retinal diseases and to assess its performance. The VGG16 and ResNet50 convolutional neural network (CNN) architectures were chosen as the standard models because of their success in medical image classification. Both models used pre-trained weights from ImageNet and the final classification layer was changed to classify the four retinal disease classes. In the training phase, only the layers specific to the task were optimized to fit the retinal fundus image database from the pre-trained models.

The classification performance of the models was compared using a comparative benchmarking analysis with the same training conditions. The accuracy, precision, recall, F1 score, ROC AUC, confusion matrix and training and validation loss were used to evaluate the performance. The benchmarking process helped in conducting a thorough comparison of the two models VGG16 and ResNet50 and this helped in identifying the best performing model for multi-class retinal disease classification.

D. Grad-CAM-Based Explainability

The best model was assessed for interpretability by applying Grad-CAM. It is an explainability technique that uses the gradients of the target class with respect to the final convolutional layer to produce heatmaps of the image region important for the class prediction. The visual explanations generated help identify clinically relevant retinal structures involved in Dry AMD, Wet AMD, moderate DR and normal fundus. Grad-CAM makes models more transparent by providing visualizations of how the model arrives at its predictions, which enables the eye doctor to check that the models are making predictions that rely on the meaningful pathological features. This enhances its reliability and clinical value for the proposed framework computer-aided retinal disease classification [20].

IV. RESULTS

A. Training and Validation Performance

The training and validation performance of the VGG16 model was done to investigate the learning behavior of the model and its generalization capability during training. The balanced retinal fundus image dataset was used to train the model and the accuracy and loss measures were monitored for each epoch. The learning curves corresponding to this indicate convergence of the model and its effectiveness in classifying the four retinal disease categories effectively.

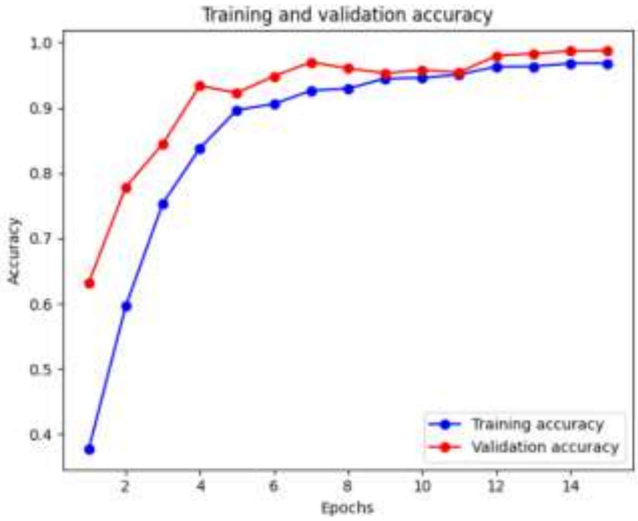

Figure 4: VGG16 Training and Validation Accuracy

Fig. 4 shows the accuracy of the training and validation data for the VGG16 model through 15 epochs. The curves depict steady growth during the training process, suggesting that the model is well-learned and converging effectively. Good generalization without any noticeable overfitting was achieved from the comparison of training and validation accuracy, making the VGG16 model appropriate for multi-class retinal disease classification.

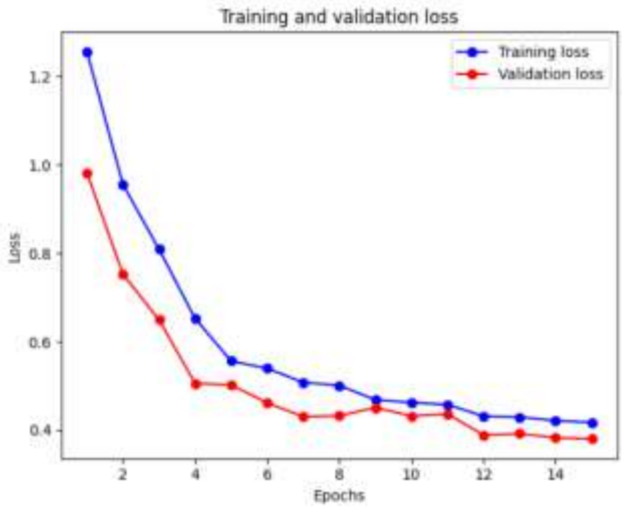

Figure 5: VGG16 Training and Validation Loss

Fig. 5 shows the training loss and validation loss of VGG16 model for 15 training epochs. It shows a clear decreasing trend for both curves, suggesting that the model is being well trained and that the curves are converging. The training and validation loss were well aligned, which shows minimal overfitting and good generalization, highlighting the effectiveness of the model to learn discriminative retinal features.

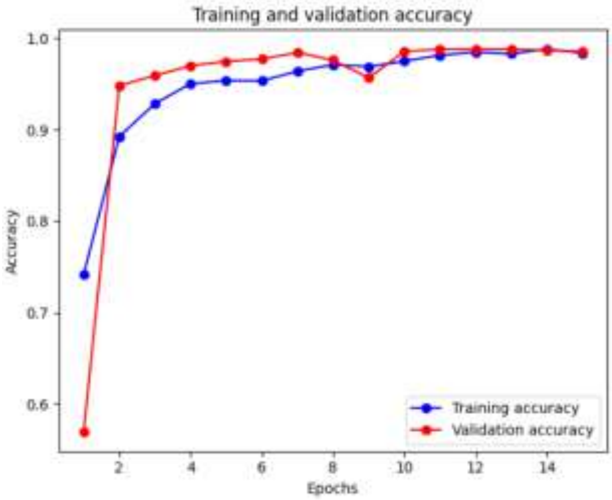

Figure 6: ResNet50 Training and Validation Accuracy

Training and validation accuracy for ResNet50 model after 15 epochs is shown in Fig. 6. The curves are consistently increasing during the training process, suggesting that the algorithm is learning the features efficiently and converging steadily. Notably, the validation accuracy was slightly higher than the training accuracy for most epochs, and both curves levelled off at the end of training, indicating that no significant overfitting occurred and this model had a good level of generalization.

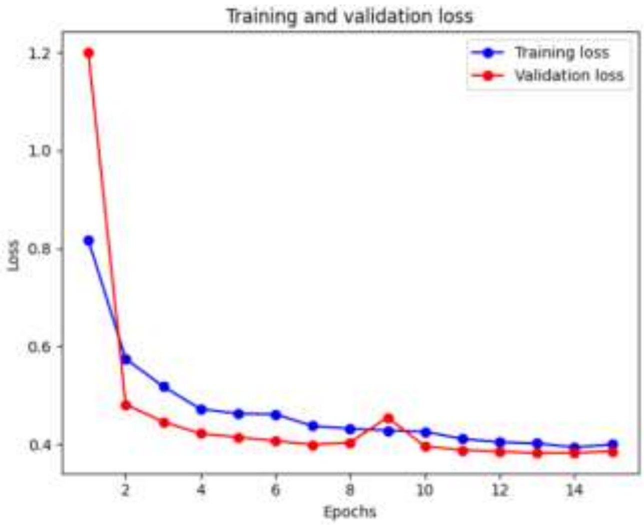

Figure 7: ResNet50 Training and Validation Loss

The training and validation loss curves of ResNet50 model over the course of 15 epochs are shown in Fig. 7. The behavior of both curves is consistent, with gradual decrease throughout the training process, which suggests effective optimization and stable convergence. The validation loss stayed slightly higher than the training loss throughout the epochs, indicating that there was a rather low level of overfitting on training the model.

B. Performance Metrics

1) Accuracy

The accuracy of the classification of the VGG16 and ResNet50 models is compared on the retinal fundus image dataset in Fig. 8.

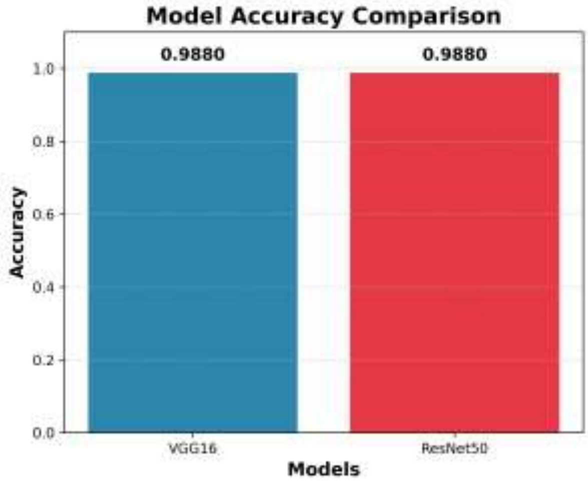

Figure 8: Accuracy Comparison of Models

The two models equally performed well with an accuracy of 98.80% for the classification of the four retinal disease categories. The high accuracy illustrates the great ability of transfer learning in learning the discriminative retinal features from fundus images. In general, both models had a high predictive power for the classification of multiple retinal diseases.

2) F1-Score

A comparison in terms of macro F1-score is shown in Fig. 9 between the VGG16 and ResNet50 models.

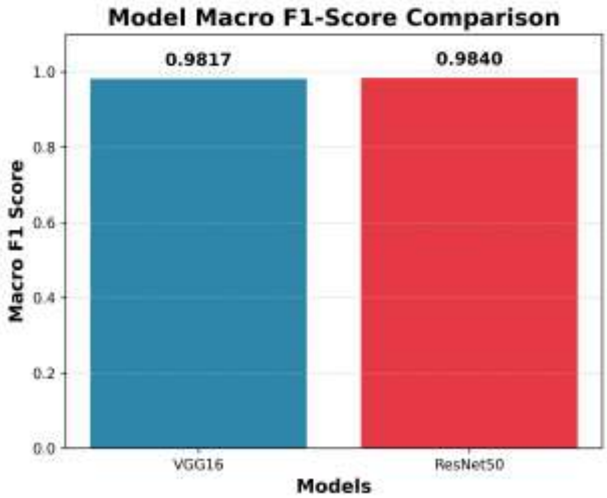

Figure 9: F1-Score Comparison of Models

VGG16 achieved the macro F1 score of 98.17% whereas ResNet50 scored slightly higher F1 score of 98.40%. The results show that both models achieve a high level of precision and recall, and have high classification reliability for the four categories of retinal disease. While the differences in performance between the models are relatively small, ResNet50 demonstrated the highest F1-score, indicating its slightly improved ability to classify retinal diseases accurately and consistently.

3) ROC-AUC

The ROC curve of the VGG16 model, across the four retinal diseases, is shown in Fig. 10.

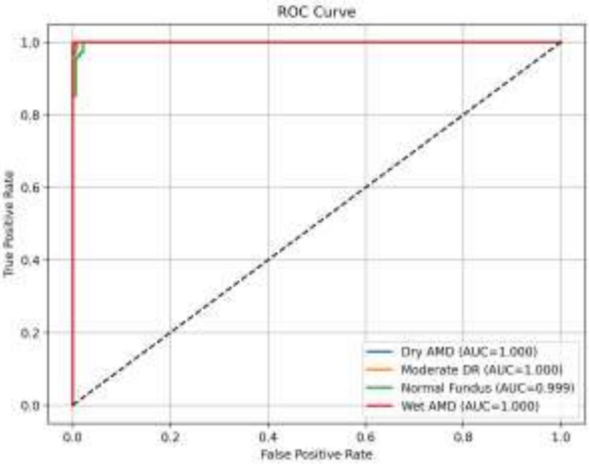

Figure 10: Multi-class ROC curves of the VGG16 model

The model performance in terms of ROC-AUC was excellent with 100% overall accuracy for the four retinal diseases. The ROC curves stay near to top left corner, showing that the true positive rate is high and the false positive rate is low. The results show that the VGG16 model can correctly classify the retinal disease classes and has excellent classification performance.

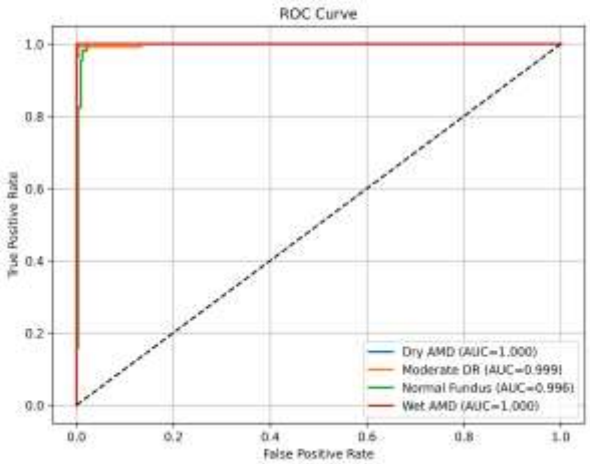

Figure 11: Multi-class ROC curves of the ResNet50 model

The ROC curves of ResNet50 model for the four retinal disease classes are shown in Fig. 11. Overall AUC score of the model was 100% with almost perfect AUC values for each class. The curves stay well away from the diagonal reference line and very close to the edge of the upper left corner, showing a good separation between disease types. The results show that ResNet50 offers robust and stable classification performance and has a good ability to classify retinal diseases in different classes.

4) Confusion Matrix

The confusion matrix of the VGG16 model on the test data is given in Fig. 12.

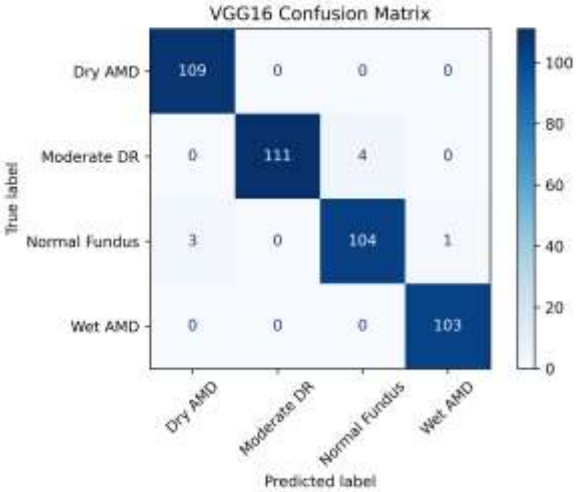

Figure 12: VGG16 Confusion Matrix

The model has successfully classified most of the retinal images with perfect classification of Dry AMD and Wet AMD. However, a few images of Moderate DR category were misclassified as Normal Fundus and a small proportion of images of the Normal Fundus category were predicted as Dry AMD and Wet AMD. The high accuracy of classification shown by the strong diagonal pattern suggests a good separation between the various retinal conditions, with minimal confusion between clinically similar conditions.

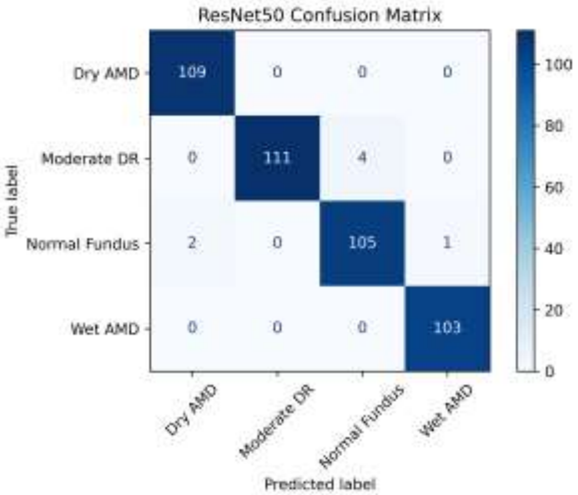

Figure 13: ResNet50 Confusion Matrix

The confusion matrix of the ResNet50 model over the test set is given in Fig. 13. The perfect classification was achieved with Dry AMD and Wet AMD and Normal Fundus was also classified with very few misclassifications. As in VGG16, most of the mispredictions were between Moderate DR and Normal Fundus, which highlights the visual likeness between early retinal changes. ResNet50 showed that its misclassification rate was further decreased for the Normal Fundus class compared with VGG16, suggesting that it can better represent the features of the class and its classification accuracy.

C. Performance Comparison of Models

The result of the overall performance of the VGG16 and ResNet50 models for multi-class retinal disease classification is shown in Table 1.

Table 1: Performance Comparison of Models

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
VGG16	98.80	0.98	0.98	98.17	1.00
ResNet50	98.80	0.98	0.98	98.40	1.00

Overall, both models showed excellent classification performance with high precision of 0.98, recall of 0.98, and ROC-AUC of 1.00 scores for each retinal disease class, demonstrating high accuracy and excellent discriminative ability. However, ResNet50 achieved a marginally better precision and recall balance by getting a slightly higher F1-score (98.40%) than VGG16 (98.17%). In general both models achieved excellent performance, ResNet50 had a slight performance advantage in classification overall because of its higher F1 score.

D. Grad-CAM-Based Model Explainability

For better transparency, to interpret and make the proposed framework more understandable, Gradient-weighted Class Activation Mapping (Grad-CAM) was used on the top performing ResNet50 model. Grad-CAM is used to obtain class-specific activation heatmaps, which pinpoint the parts of the retina that are most important to the model's prediction. The visualizations illustrate that the model will always focus on clinically relevant retinal abnormalities and not in background regions, thereby enhancing reliability and trustworthiness of the proposed computer-aided diagnostic system.

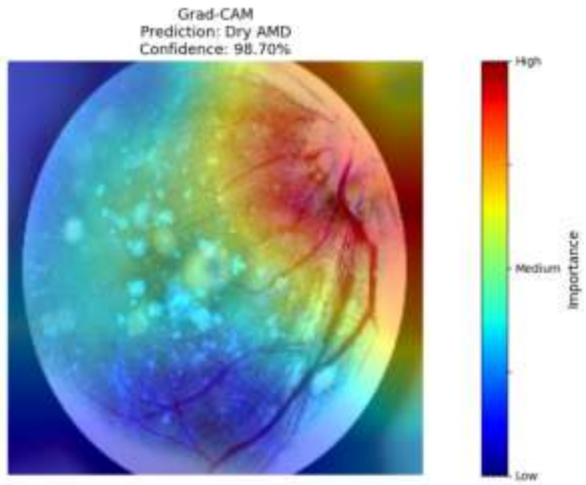

Figure 14: Explainability of the model using Grad-CAM

A sample fundus image for Dry AMD is shown with the Grad-CAM visualization in Figure 14. The heatmap is concentrated mainly in the macular area which encompasses drusen and other abnormalities of the retina associated with disease. The high importance of regions validates the clinical relevance of the model, showing that it is based on relevant clinical features instead of background features.

Overall, the Grad-CAM results indicate that the proposed ResNet50 model is learning clinically relevant retinal features for disease classification. The model's prediction is visually represented in images of retinal fundus, which aids in the interpretability of the predictions, thereby contributing to the confidence in its use as a reliable computer-aided retinal disease diagnosis system.

V. DISCUSSION

The results of the experiment prove that VGG16 and ResNet50 are able to learn discriminative retinal features for multi class classification of disease. The models were found to have an overall accuracy of 98.80%, precision of 0.98, recall of 0.98, and ROC-AUC of 1.00, demonstrating excellent classification performance and good class separability. While their performances were both very similar, ResNet50 outperformed VGG16 with a slightly higher F1-score of 98.40% versus an F1-score of 98.17% for VGG16 indicating a slight improvement in the trade-off between precision and recall. High classification performance for DR categories Dry AMD and Moderate DR suggests that these categories have distinct characteristics in their retina which allow for robust identification of the disease. Normal Fundus and Wet AMD showed high classification performance as well, which further suggests that these categories have distinct characteristics in their retina that allow for reliable identification of the disease.

The model has shown an ability to focus on clinically relevant retinal areas throughout prediction, as revealed in the Grad-CAM visualizations, further confirming the reliability of the proposed framework. The highlighted activation areas do not reflect background knowledge, but instead reflect pathology related structures in the retina, indicating that classification decisions are based on meaningful pathological features. This level of interpretability increases the trustworthiness of the proposed framework and facilitates the use of the framework as a reliable computer-aided diagnostic system.

CONCLUSION AND FUTURE WORK

This study proposed a novel deep learning approach to classify multiple retinal diseases from fundus images. The proposed method used transfer learning techniques using VGG16 and ResNet50 to classify four types of retinal disease: Dry AMD, Moderate DR, Normal Fundus and Wet AMD. Image enhancement and data balancing techniques ensured high quality images and helped in feature learning. Experimental results showed that both models resulted in an accuracy of 98.80%, a precision of 0.98, a recall of 0.98 and a ROC-AUC of 1.00. With the F1 Score, ResNet50 demonstrated 98.40% on the test set while VGG16 showed 98.17%. ResNet50 performed slightly better in terms of the balance between the precision and the recall.

The confusion matrix analysis showed that both models were able to correctly classify the most part of the retinal images with a few misclassifications between visually similar disease categories. Moreover, explainability using grad-CAM was shown to be clinically relevant with the proposed framework since it was based on clinically relevant regions of the retina with transparent and reliable predictions. In general, the proposed framework can be considered a reliable, accurate, and interpretable computer-aided diagnostic (CAD) system for screening retinal diseases. Future studies could involve the use of more comprehensive, varied retinal images, more classes of retinal disease, multimodal retinal imaging, and more sophisticated deep learning architectures for better classification performance and clinical use.

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